

Examination : End Semester Examination – May-June (2026)
 Name of the Course : B.Tech. (IT & Mathematical Innovations)
 Name of the Paper : Artificial Intelligence (DSC)
 Unique Paper Code : 3122613603
 Semester : VI
 Duration : 3 Hours
 Maximum Marks : 90

Instruction to students:

- Attempt any six questions. All questions carry equal marks.

1.

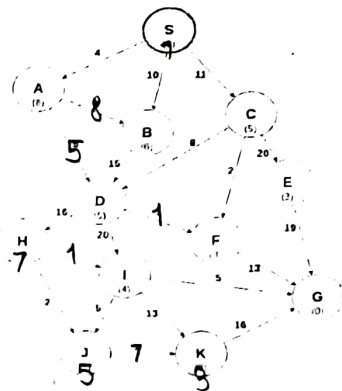
- Define PEAS for Taxi Driver. (6)
- You are given two jugs of capacity 4-gallon and 3-gallon respectively. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Write down solution by showing all intermediate states. (6)
- When do you call a machine intelligence? Name the criteria used for identifying whether a machine is intelligent or not. Explain procedure also. (3)

2.

- Using constraint satisfaction algorithm, solve the following crypt arithmetic problem. (8)

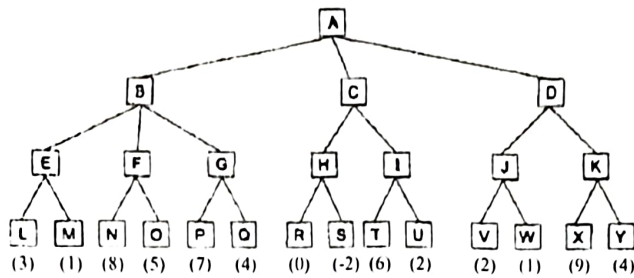
$$\begin{array}{r}
 \text{ODD} \\
 + \text{ODD} \\
 \hline
 \text{EVEN}
 \end{array}$$

- Give an example where Best First search is better than Breadth First search. (7)
3. Consider the following city map where edge values represent actual distances and vertex values represent the estimated distance to the goal city G.



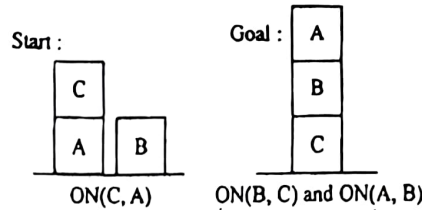
- Apply Greedy Best-First Search to find a path from S to G. Show node expansion order and total cost.
- Apply A* algorithm. Show all intermediate values and determine the optimal path.
- Compare both results and state which algorithm gives the optimal solution with justification.

4. Consider the following game tree in (assume that the first player is a maximizing player):
- What move should be chosen under min-max procedure? (7)
 - What nodes would not be needed to be examined using α - β pruning procedure? (8)



5.

- a. Consider the following block world problem and solve it using goal stack planning: (7)



- b. Consider the following axioms:

- i. Whoever can read is literate
- ii. Dolphins are not literate
- iii. Some dolphins are intelligent also
- iv. Donald is a dolphin

Using resolution to prove the "Some intelligent beings cannot read". (8)

6.

- a. Write the Conceptual Graph and FOPL representation for the following sentence: "Every bank has a locker". (5)
- b. Create a frame network for terrestrial motor vehicles (cars, trucks, motorcycles) and give one complete frame in details for cars which include the slots for the main component parts, their attributes, and relations between parts. (5)
- c. Transform the following sentence into clausal form: (5)

$$\forall x \forall y (\exists z P(x, z) \wedge P(y, z)) \rightarrow \exists u Q(x, y, u)$$

7.

- a. Identify whether the following represent classification or regression process? (5)
 - i. Predicting the price of a stock based on historical data.
 - ii. Predicting whether a credit card transaction is fraudulent or legitimate.
 - iii. Predicting the sales of a product for the next month or quarter.
 - iv. Predicting whether a customer will purchase an item or not.
 - v. Predicting the price of a house based on features such as size, location, number of rooms, etc.

- b. Consider the actual class labels and the class labels predicted by the machine learning model provided below:

Target Class	+	+	-	+	-	-	+	+	-	-	+	+
Predicted Class	+	+	+	-	+	+	-	-	-	-	+	+

Build the confusion matrix (True Positives, False Positives, True Negatives, False Negatives) and calculate metrics like accuracy, precision, recall, and F1-score from this table. (10)

• 8.

- a. Consider the following dataset that records student behaviour and whether they pass or fail.

Study Hours	Attendance	Sleep Quality	Part-time Job	Pass/Fail
Low	Poor	Bad	Yes	Fail
High	Good	Good	No	Pass
Medium	Good	Good	Yes	Pass
Low	Poor	Bad	No	Fail
High	Good	Bad	No	Pass
Medium	Poor	Good	Yes	Fail
High	Good	Good	Yes	Pass
Low	Good	Bad	Yes	Fail
Medium	Good	Good	No	Pass
High	Poor	Good	No	Pass

"Using the Naive Bayes classifier, predict whether the following student will pass or fail: <Study Hours = Medium, Attendance = Good, Sleep Quality = Good, Part-time Job = No>. (7)

- b. Consider the following sample dataset. Using the K-Nearest Neighbours (KNN) algorithm, classify an Iris flower with attributes, Sepal Length = 5.4 cm and Petal Length = 3.7 cm. Use Manhattan distance metric. Consider $k=3$ and $k=6$. (8)

Sepal Length (cm)	5.1	4.9	4.7	7	6.4	6.9	6.3	5.8	7.1	6.3
Petal Length (cm)	1.4	1.4	1.3	4.7	4.5	4.9	6	5.1	5.9	5.1
Species (Target)	Setosa	Setosa	Setosa	Versicolor	Versicolor	Versicolor	Virginica	Virginica	Virginica	Virginica

• 9.

- a. A final-year student preparing for placements can be in three states: Low (S1), Medium (S2), or High preparation (S3), with Placement (T) as a terminal state. At each state, the student can choose between Self-study or Mentorship. From S3, Self-study leads to placement with probability 0.6 (reward +10) or remains in S3 with probability 0.4 (reward +2). Mentorship leads to placement with probability 0.8 (reward +10) or remains in S3 with probability 0.2 (reward 0). Assume discount factor $\gamma = 0.9$ and $V(S3) = 8$. Formulate the MDP by specifying states, actions, rewards, and transitions. Then compute

$Q(S3, \text{Self-study})$ and $Q(S3, \text{Mentorship})$ using the Bellman equation. Based on the results, determine the best action at state S3. (8)

- b. A chess player is in a mid-game position (state S) and is considering two possible moves: Move A (aggressive attack) and Move B (defensive play). The player does not know the exact probabilities of winning from each move but has data from previous games (episodes). The outcomes are recorded as win (+100), loss (0), or draw (+50), and the discount factor is $\gamma=1$. The four observed game results are as follows:

- For Move A: Win, Loss, Win, Draw
- For Move B: Loss, Draw, Loss

Using Monte Carlo estimation, compute the action-values $Q(S,A)$ and $Q(S,B)$. Based on the computed values, determine which move is better. Further, explain whether this estimation corresponds to First-Visit or Every-Visit Monte Carlo, and justify your answer. If more game data is collected, explain how the Monte Carlo estimates will change and whether they will converge. (7)
